EXPERIMENT 10

AIM: Implementation of Hyperledger Fabric Network and Asset Management using Chaincode

THEORY:

Hyperledger Fabric is an open-source, permissioned blockchain platform designed for enterprise applications. Unlike public blockchains, it allows only authorized participants to join the network. It follows a modular architecture consisting of organizations, peers, an ordering service, a Membership Service Provider (MSP), and smart contracts known as chaincode.

A Fabric network consists of multiple organizations, each having peers that maintain a copy of the ledger. The ledger contains the blockchain, which stores transaction history, and the world state, which maintains the latest values of assets. The ordering service collects endorsed transactions, orders them into blocks, and distributes the blocks to peers. Peers validate transactions according to endorsement policies before updating the ledger.

Chaincode implements the business logic of a blockchain application. In this experiment, the Hyperledger Fabric test network was configured with two organizations, Org1 and Org2, and a channel named mychannel. The Go-based basic chaincode was deployed to the channel to perform asset-management operations.

The main asset-management operations are:
1. Create Asset: Adds a new asset with attributes such as asset ID, color, size, owner, and value.
2. Read Asset: Retrieves the details of an existing asset from the ledger.
3. Update Asset: Modifies the attributes of an existing asset.
4. Get All Assets: Retrieves the list of assets stored in the ledger.
5. Transfer Asset: Changes the ownership of an existing asset.

When a client submits a transaction, the required peers execute the chaincode and provide endorsements. The transaction is then submitted to the ordering service, which orders transactions into blocks. Peers validate the endorsements and transaction status before committing valid transactions to the ledger. In this experiment, the default endorsement policy required endorsements from both organizations, demonstrating how Fabric enforces transaction authorization and validation.

Hyperledger Fabric provides security, transparency, traceability, data consistency, and controlled access. Since transactions are recorded on the blockchain, asset changes can be tracked, making

the platform suitable for supply-chain management, inventory tracking, and enterprise asset management.

CODE AND OUTPUT:

```
b9649d3f054a: Pull complete
0b966e606d9a: Pull complete
Digest: sha256:453a0a727e82c4960b797e379adc231e853ee23ae7526db53afef77b5074117e
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-ca:1.5.17
ghcr.io/hyperledger/fabric-ca:1.5.17
===> List out hyperledger images
ghcr.io/hyperledger/fabric-peer                 2.5.16          58979b65744b   4 months ago    232MB
hyperledger/fabric-peer                         2.5             58979b65744b   4 months ago    232MB
hyperledger/fabric-peer                         2.5.16          58979b65744b   4 months ago    232MB
hyperledger/fabric-peer                         latest          58979b65744b   4 months ago    232MB
hyperledger/fabric-orderer                      2.5             1d918fc7e1e0   4 months ago    182MB
hyperledger/fabric-orderer                      2.5.16          1d918fc7e1e0   4 months ago    182MB
hyperledger/fabric-orderer                      latest          1d918fc7e1e0   4 months ago    182MB
ghcr.io/hyperledger/fabric-orderer              2.5.16          1d918fc7e1e0   4 months ago    182MB
hyperledger/fabric-ccenv                        2.5             807493f095f6   4 months ago    1.01GB
hyperledger/fabric-ccenv                        2.5.16          807493f095f6   4 months ago    1.01GB
hyperledger/fabric-ccenv                        latest          807493f095f6   4 months ago    1.01GB
ghcr.io/hyperledger/fabric-ccenv                2.5.16          807493f095f6   4 months ago    1.01GB
hyperledger/fabric-baseos                       2.5             109b4a15b282   4 months ago    257MB
hyperledger/fabric-baseos                       2.5.16          109b4a15b282   4 months ago    257MB
hyperledger/fabric-baseos                       latest          109b4a15b282   4 months ago    257MB
ghcr.io/hyperledger/fabric-baseos               2.5.16          109b4a15b282   4 months ago    257MB
hyperledger/fabric-ca                           1.5             453a0a727e82   8 months ago    381MB
hyperledger/fabric-ca                           1.5.17          453a0a727e82   8 months ago    381MB
hyperledger/fabric-ca                           latest          453a0a727e82   8 months ago    381MB
ghcr.io/hyperledger/fabric-ca                   1.5.17          453a0a727e82   8 months ago    381MB
hyperledger/fabric-peer                         2.5             58979b65744b   4 months ago    232MB
hyperledger/fabric-peer                         2.5.16          58979b65744b   4 months ago    232MB
hyperledger/fabric-peer                         latest          58979b65744b   4 months ago    232MB
ghcr.io/hyperledger/fabric-peer                 2.5.16          58979b65744b   4 months ago    232MB
hyperledger/fabric-orderer                      2.5             1d918fc7e1e0   4 months ago    182MB
hyperledger/fabric-orderer                      2.5.16          1d918fc7e1e0   4 months ago    182MB
hyperledger/fabric-orderer                      latest          1d918fc7e1e0   4 months ago    182MB
ghcr.io/hyperledger/fabric-orderer              2.5.16          1d918fc7e1e0   4 months ago    182MB
hyperledger/fabric-ccenv                        2.5             807493f095f6   4 months ago    1.01GB
hyperledger/fabric-ccenv                        2.5.16          807493f095f6   4 months ago    1.01GB
hyperledger/fabric-ccenv                        latest          807493f095f6   4 months ago    1.01GB
ghcr.io/hyperledger/fabric-ccenv                2.5.16          807493f095f6   4 months ago    1.01GB
hyperledger/fabric-baseos                       2.5             109b4a15b282   4 months ago    257MB
hyperledger/fabric-baseos                       2.5.16          109b4a15b282   4 months ago    257MB
hyperledger/fabric-baseos                       latest          109b4a15b282   4 months ago    257MB
```

```
chmod +x install-fabric.sh

./install-fabric.sh docker samples binary

docker images | grep fabric
Cloning into 'fabric-samples'...
remote: Enumerating objects: 16063, done.
remote: Counting objects: 100% (3470/3470), done.
remote: Compressing objects: 100% (641/641), done.
remote: Total 16063 (delta 3045), reused 2829 (delta 2829), pack-reused 12593 (from 3)
Receiving objects: 100% (16063/16063), 24.08 MiB | 590.00 KiB/s, done.
Resolving deltas: 100% (9276/9276), done.

Clone hyperledger/fabric-samples repo

==> Already in fabric-samples repo
fabric-samples v2.5.16 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric

Pull Hyperledger Fabric binaries

===> Downloading version 2.5.16 platform specific fabric binaries
===> Downloading:  https://github.com/hyperledger/fabric/releases/download/v2.5.16/hyperledger-fabric-linux-amd64-2.5.16.tar.gz
===> Will unpack to: /home/shravanya/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100  121M  100  121M    0     0   792k      0  0:02:37  0:02:37 --:--:--  593k
==> Done.
===> Downloading version 1.5.17 platform specific fabric-ca-client binary
===> Downloading:  https://github.com/hyperledger/fabric-ca/releases/download/v1.5.17/hyperledger-fabric-ca-linux-amd64-1.5.17.tar.gz
===> Will unpack to: /home/shravanya/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:--  0:00:01 --:--:--     0
100 31.6M  100 31.6M    0     0   862k      0  0:00:37  0:00:37 --:--:--  649k
==> Done.
```

```
shravanya@BOOK-HCO17TJGP2:~/fabric-samples/test-network$ cd ~/fabric-samples/test-network

echo "===== DOCKER CONTAINERS ====="
docker ps -a

echo "===== CHANNEL / NETWORK STATUS ====="
docker network ls

echo "===== ORDERER LOGS ====="
docker logs --tail 60 orderer.example.com

echo "===== ORG1 PEER LOGS ====="
docker logs --tail 60 peer0.org1.example.com
===== DOCKER CONTAINERS =====
CONTAINER ID   IMAGE                                          COMMAND                  CREATED         STATUS                       PORTS                                                
```

```
er peers in the organization channel=mychannel
2026-10-26 06:34:49.368 UTC 002f INFO [endorser] callChaincode -> finished chaincode: cscc duration: 61ms channel= txID=f33c1a2c
2026-10-26 06:34:49.385 UTC 0031 INFO [peer.blocksprovider] DeliverBlocks -> Pulling next blocks from ordering service channel=mychannel orderer-address=ord
erer.example.com:7050 nextBlock=1
2026-10-26 06:37:38.994 UTC 0032 INFO [ledgermgmt] CreateLedger -> Creating ledger [mychannel] with genesis block
2026-10-26 06:37:38.994 UTC 0033 INFO [endorser] callChaincode -> finished chaincode: cscc duration: 4ms channel= txID=9add236a
2026-10-26 06:37:42.150 UTC 0035 INFO [ledgermgmt] CreateLedger -> Creating ledger [mychannel] with genesis block
2026-10-26 06:37:42.150 UTC 0036 INFO [endorser] callChaincode -> finished chaincode: cscc duration: 12ms channel= txID=11a07923
2026-10-26 06:37:45.320 UTC 0038 INFO [ledgermgmt] CreateLedger -> Creating ledger [mychannel] with genesis block
2026-10-26 06:37:45.320 UTC 0039 INFO [endorser] callChaincode -> finished chaincode: cscc duration: 4ms channel= txID=b2e19d05
2026-10-26 06:37:48.439 UTC 003b INFO [ledgermgmt] CreateLedger -> Creating ledger [mychannel] with genesis block
2026-10-26 06:37:48.440 UTC 003c INFO [endorser] callChaincode -> finished chaincode: cscc duration: 3ms channel= txID=706be0a8
===== ORDERER ERRORS =====
        General.ConnectionTimeout = 0s
2026-10-26 06:34:52.544 UTC 002d WARN [common.deliver] Handle -> Error reading from 172.22.0.1:33264: rpc error: code = Canceled desc = context canceled
2026-10-26 06:34:52.544 UTC 002e INFO [comm.grpc.server] 1 -> streaming call completed grpc.service=orderer.AtomicBroadcast grpc.method=Deliver grpc.peer_ad
dress=172.22.0.1:33264 error="rpc error: code = Canceled desc = context canceled" grpc.code=Canceled grpc.call_duration=6.078898ms
===== DOCKER NETWORK =====
NETWORK ID     NAME                          DRIVER    SCOPE
f7fa582bc5e9   bridge                        bridge    local
b829aba98f08   fabric_test                   bridge    local
fe4bbd0beb40   host                          host      local
5bd7fd8838b5   mediawiki-net                 bridge    local
de5c642e23a8   medication-backend_default    bridge    local
b7fbd6307c43   none                          null      local
91c76c21af1b   snipeit-docker_default        bridge    local
c7de6aa645c9   zabbix-net                    bridge    local
90efcc6122fe   zammad_default                bridge    local
peer0.org1.example.com orderer.example.com peer0.org2.example.com
===== REQUIRED TOOLS =====
jq-1.6
peer:
 Version: v2.5.16
 Commit SHA: f871cf9
 Go version: go1.26.4
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger
```

```
                                                                                                                                                hungry_elgamal
Using docker and docker-compose
Generating channel genesis block 'mychannel.block'
Using organization 1
/home/shravanya/fabric-samples/test-network/../bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-10-26 12:10:46.700 IST 0001 INFO [common.tools.configtxgen] main -> Loading configuration
2026-10-26 12:10:46.703 IST 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> orderer type: etcdraft
2026-10-26 12:10:46.703 IST 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization -> Orderer.EtcdRaft.Options unset, setting to tick_inter
val:"500ms" election_tick:10 heartbeat_tick:1 max_inflight_blocks:5 snapshot_interval_size:16777216
2026-10-26 12:10:46.703 IST 0004 INFO [common.tools.configtxgen.localconfig] Load -> Loaded configuration: /home/shravanya/fabric-samples/test-network/confi
gtx/configtx.yaml
2026-10-26 12:10:46.705 IST 0005 INFO [common.tools.configtxgen] doOutputBlock -> Generating genesis block
2026-10-26 12:10:46.705 IST 0006 INFO [common.tools.configtxgen] doOutputBlock -> Creating application channel genesis block
2026-10-26 12:10:46.706 IST 0007 INFO [common.tools.configtxgen] doOutputBlock -> Writing genesis block
+ res=0
Creating channel mychannel
Adding orderers
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 201
{
        "name": "mychannel",
        "url": "/participation/v1/channels/mychannel",
        "consensusRelation": "consenter",
        "status": "active",
        "height": 1
}

Channel 'mychannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0
2026-10-26 12:10:52.895 IST 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-10-26 12:10:52.986 IST 0002 INFO [channelCmd] executeJoin -> Successfully submitted proposal to join channel
Joining org2 peer to the channel...
Using organization 2
```

```
Hit:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  golang-1.18-go golang-1.18-src golang-src
Suggested packages:
  bzr | brz mercurial subversion
The following NEW packages will be installed:
  golang-1.18-go golang-1.18-src golang-go golang-src
0 upgraded, 4 newly installed, 0 to remove and 258 not upgraded.
Need to get 82.4 MB of archives.
After this operation, 437 MB of additional disk space will be used.
Ign:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 golang-1.18-src all 1.18.1-1ubuntu1.3
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 golang-1.18-go amd64 1.18.1-1ubuntu1.3 [66.1 MB]
Ign:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 golang-1.18-go amd64 1.18.1-1ubuntu1.3
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 golang-src all 2:1.18~0ubuntu2 [4438 B]
Get:4 http://archive.ubuntu.com/ubuntu jammy/main amd64 golang-go amd64 2:1.18~0ubuntu2 [41.8 kB]
Ign:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 golang-1.18-src all 1.18.1-1ubuntu1.3
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 golang-1.18-go amd64 1.18.1-1ubuntu1.3 [66.1 MB]
Ign:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 golang-1.18-go amd64 1.18.1-1ubuntu1.3
Ign:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 golang-1.18-src all 1.18.1-1ubuntu1.3
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 golang-1.18-go amd64 1.18.1-1ubuntu1.3 [66.1 MB]
Ign:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 golang-1.18-src all 1.18.1-1ubuntu1.3
Err:1 http://security.ubuntu.com/ubuntu jammy-updates/main amd64 golang-1.18-src all 1.18.1-1ubuntu1.3
  500  Internal Server Error [IP: 91.189.91.83 80]
Fetched 66.2 MB in 3min 37s (305 kB/s)
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/g/golang-1.18/golang-1.18-src_1.18.1-1ubuntu1.3_all.deb
89.91.83 80]
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
===== GO VERSION =====
Command 'go' not found, but can be installed with:
sudo snap install go         # version 1.26.2, or
sudo apt  install golang-go  # version 2:1.18~0ubuntu2
sudo apt  install gccgo-go   # version 2:1.18~0ubuntu2
See 'snap info go' for additional versions.
===== GO LOCATION =====
```

```
shravanya@BOOK-HCO17TJGP2:~/fabric-samples/test-network$ cd ~/fabric-samples/test-network

echo "===== GO ====="
go version

echo "===== NETWORK CONTAINERS ====="
docker ps --format "table {{.Names}}\t{{.Status}}"

echo "===== COMMITTED CHAINCODE ====="
peer lifecycle chaincode querycommitted \
  --channelID mychannel

echo "===== INSTALLED CHAINCODE ====="
peer lifecycle chaincode queryinstalled
===== GO =====
Command 'go' not found, but can be installed with:
sudo snap install go         # version 1.26.2, or
sudo apt  install golang-go  # version 2:1.18~0ubuntu2
sudo apt  install gccgo-go   # version 2:1.18~0ubuntu2
See 'snap info go' for additional versions.
===== NETWORK CONTAINERS =====
NAMES                            STATUS
peer0.org2.example.com           Up 6 minutes
peer0.org1.example.com           Up 6 minutes
orderer.example.com              Up 6 minutes
snipeit-app                      Up 31 minutes
snipeit-db                       Up 31 minutes
zammad-zammad-nginx-1            Up 31 minutes
zammad-zammad-railsserver-1      Up 31 minutes
zammad-zammad-backup-1           Up 31 minutes
zammad-zammad-websocket-1        Up 31 minutes
zammad-zammad-scheduler-1        Up 31 minutes
zammad-zammad-postgresql-1       Up 31 minutes
zammad-zammad-redis-1            Up 31 minutes
zammad-zammad-memcached-1        Up 31 minutes
zammad-zammad-elasticsearch-1    Up 31 minutes
===== COMMITTED CHAINCODE =====
Committed chaincode definitions on channel 'mychannel':
===== INSTALLED CHAINCODE =====
Installed chaincodes on peer:
```

```
Location: https://dl.google.com/go/go1.25.0.linux-amd64.tar.gz [following]
--2026-10-28 08:21:00--  https://dl.google.com/go/go1.25.0.linux-amd64.tar.gz
Resolving dl.google.com (dl.google.com)... 142.251.126.190, 142.251.126.136, 142.251.126.93, ...
Connecting to dl.google.com (dl.google.com)|142.251.126.190|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 59659213 (57M) [application/x-gzip]
Saving to: ‘go1.25.0.linux-amd64.tar.gz’

go1.25.0.linux-amd64.tar.gz   100%[===============================================>]  56.89M  9.28MB/s    in 6.2s

2026-10-28 08:21:06 (9.12 MB/s) - ‘go1.25.0.linux-amd64.tar.gz’ saved [59659213/59659213]

go version go1.25.0 linux/amd64
shravanya@BOOK-HCO17TJGP2:/tmp$ cd ~/fabric-samples/test-network

export PATH="/usr/local/go/bin:$HOME/fabric-samples/bin:/usr/bin:$PATH"
export FABRIC_CFG_PATH="$HOME/fabric-samples/config"

./network.sh deployCC \
  -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-go \
  -ccl go
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-go
- CC_SRC_LANGUAGE: go
- CC_VERSION: 1.0
```

```
~/fabric-samples/test-network
Finished vendoring Go dependencies
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basic/chaincode-go --lang golang --label basic_
1.0
+ res=0
Chaincode is packaged
Installing chaincode on peer0.org1...
Using organization 1
+ peer lifecycle chaincode queryinstalled --output json
+ jq -r 'try (.installed_chaincodes[].package_id)'
+ grep '^basic_1.0:d0b27e1af09f3039dabac82678c64e8807311f9405008efb867ba8895dd643aa$'
+ test 1 -ne 0
+ peer lifecycle chaincode install basic.tar.gz
+ res=0
2026-10-28 08:21:40.101 IST 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal -> Installed remotely: response:<s
tatus:200 payload:"\nJbasic_1.0:d0b27e1af09f3039dabac82678c64e8807311f9405008efb867ba8895dd643aa\022\tbasic_1.0" >
2026-10-28 08:21:40.102 IST 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal -> Chaincode code package identifi
er: basic_1.0:d0b27e1af09f3039dabac82678c64e8807311f9405008efb867ba8895dd643aa
Chaincode is installed on peer0.org1
Install chaincode on peer0.org2...
Using organization 2
+ peer lifecycle chaincode queryinstalled --output json
+ jq -r 'try (.installed_chaincodes[].package_id)'
+ grep '^basic_1.0:d0b27e1af09f3039dabac82678c64e8807311f9405008efb867ba8895dd643aa$'
+ test 1 -ne 0
+ peer lifecycle chaincode install basic.tar.gz
```

```
327a12d9886bedde3] committed with status (VALID) at localhost:9051
Chaincode definition committed on channel 'mychannel'
Using organization 1
Querying chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to Query committed status on peer0.org1, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org2 on channel 'mychannel'
Chaincode initialization is not required
shravanya@BOOK-HCO17TJGP2:~/fabric-samples/test-network$ cd ~/fabric-samples/test-network

export PATH="/usr/local/go/bin:$HOME/fabric-samples/bin:/usr/bin:$PATH"
export FABRIC_CFG_PATH="$HOME/fabric-samples/config"

peer lifecycle chaincode querycommitted \
  --channelID mychannel \
  --name basic
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
```

```
shravanya@BOOK-HCO17TJGP2:~/fabric-samples/test-network$ export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export CORE_PEER_TLS_ROOTCERT_FILE="$PWD/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt"
export CORE_PEER_MSPCONFIGPATH="$PWD/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp"
export CORE_PEER_ADDRESS=localhost:7051

peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile "$PWD/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem" \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
  -c '{"function":"CreateAsset","Args":["asset10","blue","5","Shravanya Andhale","500"]}'
2026-10-28 08:28:02.961 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
```

```
shravanya@BOOK-HCO17TJGP2:~/fabric-samples/test-network$ peer chaincode query \
  -C mychannel \
  -n basic \
  -c '{"function":"ReadAsset","Args":["asset10"]}'
{"AppraisedValue":500,"Color":"blue","ID":"asset10","Owner":"Shravanya Andhale","Size":5}
```

```
shravanya@BOOK-HCO17TJGP2:~/fabric-samples/test-network$ peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile "$PWD/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem" \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
  -c '{"function":"UpdateAsset","Args":["asset10","green","10","Shravanya Andhale","750"]}'
2026-10-28 08:31:26.092 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
```

```
shravanya@BOOK-HCO17TJGP2:~/fabric-samples/test-network$ peer chaincode query \
  -C mychannel \
  -n basic \
  -c '{"function":"ReadAsset","Args":["asset10"]}'
{"AppraisedValue":750,"Color":"green","ID":"asset10","Owner":"Shravanya Andhale","Size":10}
```

```
shravanya@BOOK-HCO17TJGP2:~/fabric-samples/test-network$ peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile "$PWD/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.exampl
e.com-cert.pem" \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
  -c '{"function":"TransferAsset","Args":["asset10","Manu Andhale"]}'
2026-10-28 08:32:52.412 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: stat
us:200 payload:"Shravanya Andhale"
```

```
shravanya@BOOK-HCO17TJGP2:~/fabric-samples/test-network$ peer chaincode query \
  -C mychannel \
  -n basic \
  -c '{"function":"ReadAsset","Args":["asset10"]}'
{"AppraisedValue":750,"Color":"green","ID":"asset10","Owner":"Manu Andhale","Size":10}
```

```
shravanya@BOOK-HCO17TJGP2:~/fabric-samples/test-network$ cd ~/fabric-samples/test-network

export PATH="/usr/local/go/bin:$HOME/fabric-samples/bin:/usr/bin:$PATH"
export FABRIC_CFG_PATH="$HOME/fabric-samples/config"

export CORE_PEER_LOCALMSPID="Org1MSP"
export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_TLS_ROOTCERT_FILE="$PWD/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/t
ls/ca.crt"
export CORE_PEER_MSPCONFIGPATH="$PWD/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp"
export CORE_PEER_ADDRESS=localhost:7051

echo "===== ALL ASSETS ====="
peer chaincode query -C mychannel -n basic \
  -c '{"function":"GetAllAssets","Args":[]}'

echo "===== ASSET10 ====="
peer chaincode query -C mychannel -n basic \
  -c '{"function":"ReadAsset","Args":["asset10"]}'
===== ALL ASSETS =====
[{"AppraisedValue":750,"Color":"green","ID":"asset10","Owner":"Manu Andhale","Size":10}]
===== ASSET10 =====
{"AppraisedValue":750,"Color":"green","ID":"asset10","Owner":"Manu Andhale","Size":10}
```

CONCLUSION:

Thus, the Hyperledger Fabric test network was configured, and the Go-based chaincode was deployed on mychannel to demonstrate blockchain-based asset management. The experiment illustrates how assets can be created, read, updated, listed, and transferred through chaincode while following the network's endorsement policy. It demonstrates the use of a permissioned blockchain for secure, traceable, and reliable asset management.